

Predicting Progression from Well-Differentiated to Dedifferentiated Liposarcoma

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Introduction

- Liposarcomas arise from adipocytes ubiquitously in the body.
- Well-differentiated liposarcoma (WDLPS) accounts for approximately 50% of all cases.
- Dedifferentiated liposarcoma (DDLPS) is more aggressive and develops from well-differentiated tumors.
- WDLPS can repeatedly recur and invade locally but cannot metastasize unless it dedifferentiates.
- WDLPS and DDLPS primarily amplify MDM2 (suppresses p53 pathway) and CDK4 (promotes cell cycle progression) on chromosome 12q13-15¹.
- When WDLPS dedifferentiates (~10% all cases), slow-growing fat-like tumor changes to more aggressive, high-grade cancer with less mature fat features.
- Approximately 90% of DDLPS cases *de novo*, but studies demonstrate many may be misinterpretations due to hidden WDLPS components¹.
- Risk of dedifferentiation increases with each subsequent recurrence⁴.
- Retroperitoneal and paratesticular WDLPS carry up to 20% higher risk of dedifferentiation compared to limb WDLPS².

Motivation

Predicting whether a well-differentiated tumor will dedifferentiate is important for early detection and treatment while it is still lower-grade, unable to metastasize, and has a higher survivability³.

Methods and Dataset

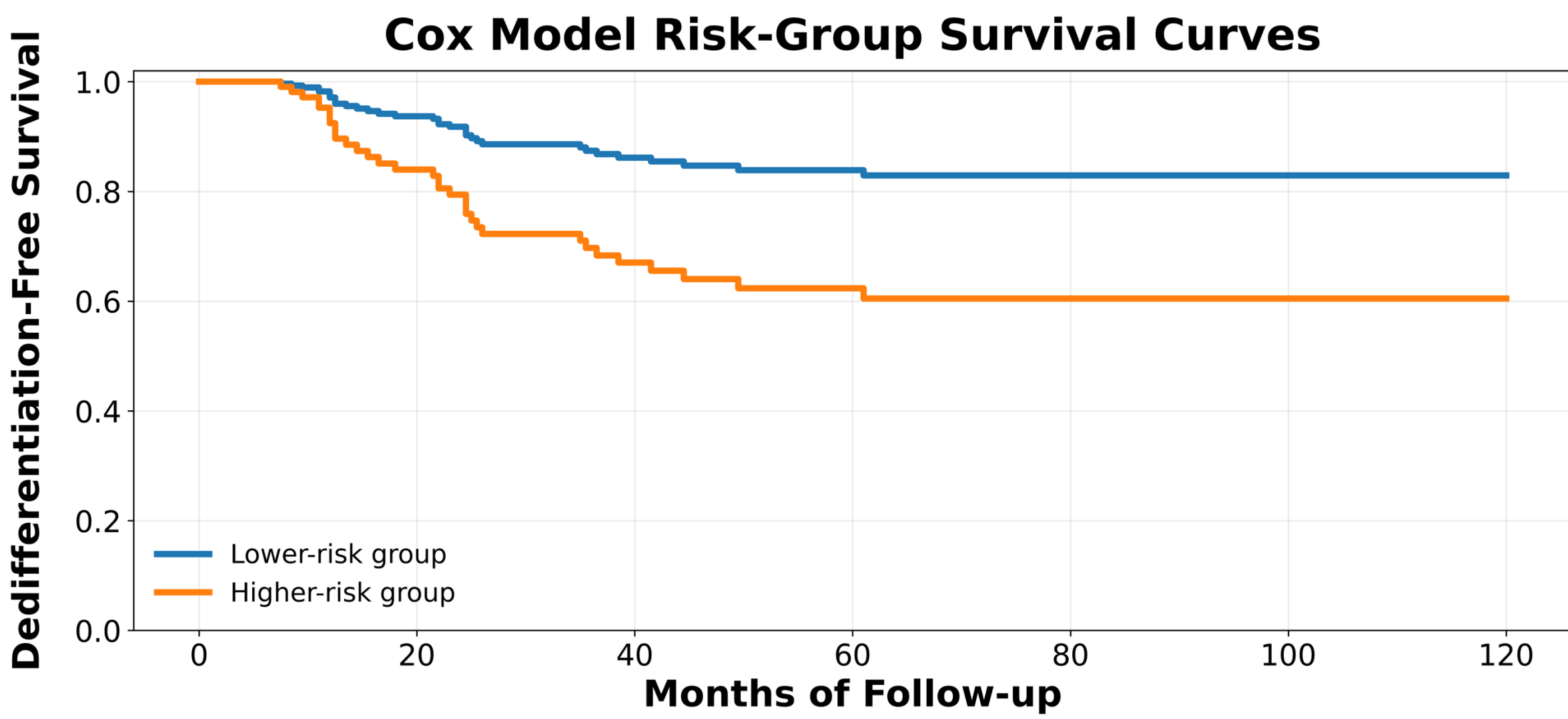
Two-step approach for evaluating the predictors of dedifferentiation:

- Multivariate Cox proportional hazards regression to evaluate statistically significant factors, 5-fold cross validation over ten iterations.
- Logistic regression to observe the predictive power of the strongest features, out-of-fold cross validation.

The cohort data was synthetically generated based on the population statistics of Grady Health System.

Stable WDLPS	DDLPS	Total	Median Follow-Up
237	63	300	51.4 months

Results



Best Predictors for Disease Progression, Cox Hazards

Predictor	Scale	Adjusted HR	95% CI	P value
CDK4 copy number	Per 1-SD increase	1.51	1.11–2.06	0.009
Genomic complexity	Per 1-SD increase	1.32	1.04–1.69	0.022
Suspicious nodular focus	Presence vs absence	1.86	1.05–3.29	0.033

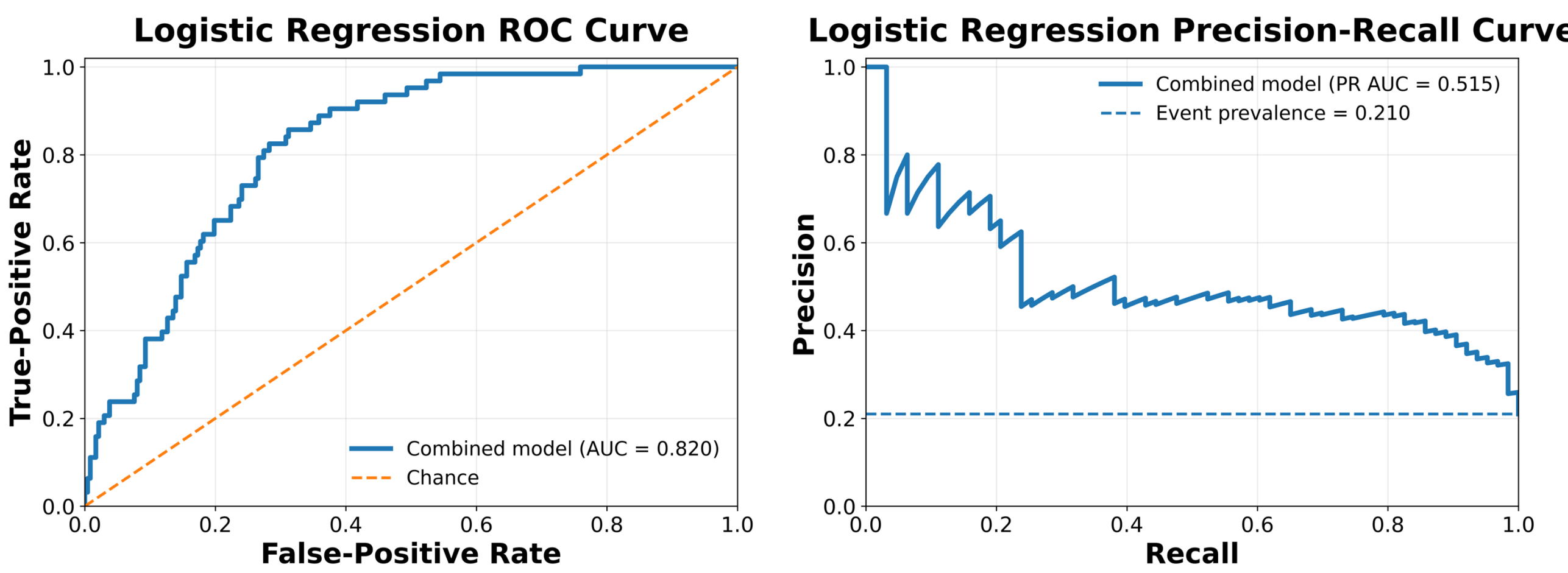
C-index 0.729 ± 0.120

CDK4 copy number, genomic complexity, and suspicious nodular focus were statistically significant for Cox hazards.

Performance of Disease Progression Prediction by Logistic Regression

ROC AUC	PRC AUC	Precision	Recall	F1
0.820	0.515	0.435	0.794	0.562

Hyperparameters: L2 regularization, C = 0.03



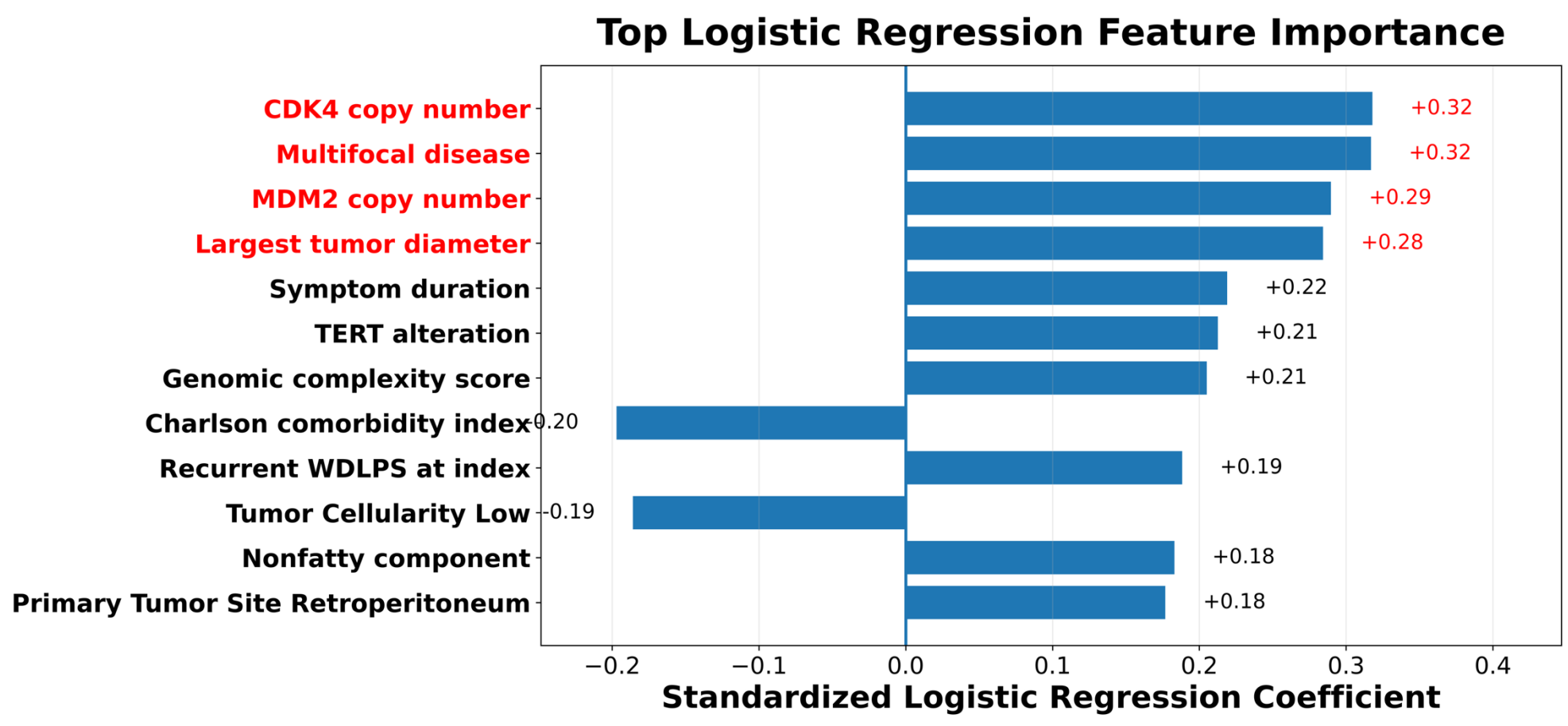
Confusion Matrix for Disease Progression Prediction by Logistic Regression

	Predicted Stable WDLPS	Predicted DDLPS	Total
Stable WDLPS	172	65	227
DDLPS	13	50	63
Total	185	115	300

The model demonstrates a relatively strong negative predictive performance but struggles with precision, misclassifying WDLPS cases as DDLPS.

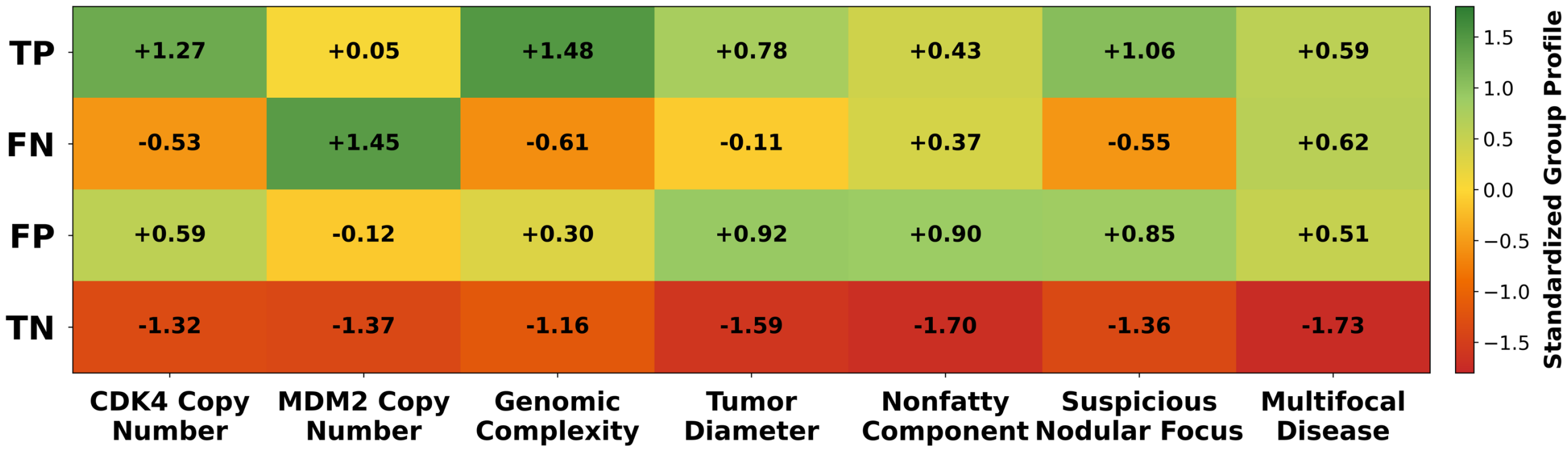
Analysis

Feature Importance for Disease Progression Prediction by Logistic Regression



CDK4 copy number, multifocal disease, MDM2 copy number, and largest tumor diameter were the strongest features in the model. Tumor cellularity low and Charlson Comorbidity Index had an inverse effect on the model.

Disease Progression Prediction Error Profile



Higher CDK4 copy number, multifocal disease, and larger tumor diameter were associated with false positive errors, while multifocal disease was associated with false negative errors. Higher MDM2 copy numbers carried an exceptionally strong association to false negatives.

False negatives had a lower CDK4 copy number, lower genomic complexity, and less suspicious nodular findings than true positives. False positives had very similar profiles to true positives, which could be explained by tumors not being dedifferentiated at the time of follow-up. Drop-out analysis based on recurrence and time criteria could reveal more about which factors influence false positive and true positive predictions.

Limitations

While accounted for here by exploiting cross validation, the small size of the dataset prevents training and evaluation of a more complex predictive model. As a result, the final model is not time-sensitive, which is an important part of predictive disease modeling. The limitation of the patient population to Grady Health System statistics prevents general application to other institutions.

Conclusions

- Cox survival model shows CDK4 copy number, genomic complexity, suspicious nodular focus demonstrate significantly strong predictors.
- Logistic regression presents a promising sensitive predictive model, but poor positive predictive value.
- Future directions:
 - Generalization to broader patient population through evaluation on external hospital data.
 - Design time-sensitive predictive model to address limitations of logistic regression.

Additional Materials

References

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[2] Weiss, S. W., & Rao, V. K. (1992). Well-differentiated liposarcoma (atypical lipoma) of deep soft tissue of the extremities, retroperitoneum, and miscellaneous sites: A follow-up study of 92 cases with analysis of the incidence of "dedifferentiation." *The American Journal of Surgical Pathology*, 16(11), 1051–1058. <https://doi.org/10.1097/00000478-199211000-00003>

[3] Serguienko, A., Braadland, P., Meza-Zepeda, L. A., Bjerkehagen, B., & Myklebost, O. (2020). Accurate 3-gene-signature for early diagnosis of liposarcoma progression. *Clinical Sarcoma Research*, 10, Article 4. <https://doi.org/10.1186/s13569-020-0126-1>

[4] Crago, A. M., & Singer, S. (2011). Clinical and molecular approaches to well-differentiated and dedifferentiated liposarcoma. *Current Opinion in Oncology*, 23(4), 373–378. <https://doi.org/10.1097/CCO.0b013e32834796e6>

Appendix